

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. Branch : E&TC

Semester : VIII

Subject Code & Name: BTETPE802F Digital IC Design

Max Marks: 60

Date:07/07/2022

Duration: 3.45 Hr.

Instructions to the Students:

1. All Questions are Compulsory
2. Draw neat diagram wherever necessary.
3. Figures to right indicates full marks
4. Assume suitable data wherever necessary and mention it clearly

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		
A) Explain various short channel effects of MOSFET	2	06
B) Implement binary to Gray code converter using static CMOS logic	3	06
C) What are the drawbacks of dynamic logic? How they are overcome in Domino logic	4	06
Q.2 Solve Any Two of the following.		
A) Write a short note on Mirror adder	1	06
B) Implement 4:1 multiplexer using static CMOS logic. How many transistors are required?	4	06
C) Implement $Y=AB+CD$ using static CMOS logic family	3	06
Q. 3 Solve Any Two of the following.		
A) Find the logical effort required for 2 input XOR gate with equal rise & fall time	2	06
B) Explain Body effect on MOSFET	1	06
C) Write a short note on square root adder	1	06
Q.4 Solve Any Two of the following.		
A) What are skewed and unskewed gates? Draw two input NOR skewed and unskewed gate	3	06
B) Explain Set up and Hold time measurement for D Flip Flop	3	06
C) Implement $Y=ABC$ using CMOS dynamic logic	4	06
Q. 5 Solve Any Two of the following.		
A) Explain Channel Length Modulation the non-ideal effect of MOS transistor	1	06
B) Draw CMOS inverter's VI characteristics & explain them	2	06
C) Calculate logical effort for three input NOR gate with equal rise and fall time.	2	06

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